

TRADING & MARKET ANALYTICS PORTFOLIO PROJECT

Permian–Waha Basis

Where the Permian discount widens — and what a trading desk should do when it does

THE SITUATION

A Gas & Power trading desk needs an early-warning view of Permian basis risk

Waha (the Permian hub) regularly trades at a discount to Henry Hub, the national benchmark. When that discount widens or reverses unexpectedly, it matters for hedge timing, pipeline-capacity nominations, and positioning — but only if the desk can tell a genuine dislocation from Waha's normal, everyday discount.

THE QUESTION

When does Waha decouple from Henry Hub enough to actually matter for a trading decision?

Data & approach

Henry Hub

FRED API, daily. EIA's own route for this stopped updating in April 2024 — confirmed by inspecting it directly, then switched sources.

Waha & rig count

No clean API exists for Waha anywhere. Regex-parsed from EIA Weekly Update report prose — the genuine messy-data problem of this project.

Storage

EIA API, South Central region specifically (not the Lower 48 aggregate) — the figure actually relevant to Permian dynamics.

Method: basis = Waha – Henry Hub, weekly. Anomalies flagged with a rolling z-score ($|z| > 2$, trailing 12-observation window).

FINDING 1 — THE PERSISTENT DISCOUNT

—\$0.91

average weekly basis, 2022–2026 (median —\$0.63/MMBtu)

Waha traded below Henry Hub in the large majority of 111 usable weeks — consistent with known Permian Basin takeaway-capacity constraints.

FINDING 2 — A LONG NEGATIVE TAIL

Most weeks are a modest discount. A few are severe.

MIN	25TH PCT	MEDIAN	75TH PCT	MAX
-\$5.93	-\$1.17	-\$0.63	-\$0.30	+\$2.01

The 75th percentile is still only $-\$0.30/\text{MMBtu}$ — but the tail runs all the way to a $\$5.93$ collapse. Dislocations are infrequent, and severe when they happen.

FINDING 3 — STATISTICALLY UNUSUAL WEEKS

9 of 111

weeks flagged as unusual (Iz-scoreI > 2 vs. a trailing 12-observation mean) — two distinct patterns, not one

Week	Basis	Pattern
Oct 26, 2022	−\$5.93	Deep dislocation
Aug 28, 2024	−\$5.56	Deep dislocation
Mar 19, 2025	−\$4.78	Deep dislocation
Dec 13, 2023	+\$1.06	Positive anomaly
Dec 18, 2024	+\$2.01	Largest positive anomaly

THE DIAGNOSIS — WHY

The deep dislocations

Oct 2022, Aug 2024, and Mar 2025 align with a widely reported 2022–2024 period of chronic Permian oversupply and pipeline maintenance. Directly confirmed in EIA's own commentary for August 2024; the other two are consistent with this broader trend, not individually source-verified.

This is a plausible, evidence-adjacent hypothesis for the December reversal — not a confirmed cause. The exact EIA report for that week could not be located to state a cause directly.

The Dec 2024 reversal

Coincides with the 2.5 Bcf/d Matterhorn Express Pipeline adding Permian takeaway capacity starting October 2024 — weeks before the anomaly. Weather data for that week was near-normal, ruling out a demand-spike explanation.

CAN STORAGE OR RIG COUNT PREDICT THIS?

Tested rigorously — one clear no, one real but unconfirmed

South Central storage

No relationship

Every lag tested (0/1/2/4/8 weeks) is statistically non-significant, $p > 0.28$ throughout. A real negative finding, not an inconclusive one.

Rig count

$r \approx -0.24$ to -0.29

Consistent at 0–4 week lags ($n = 67–71$), nominally $p < 0.05$ — but fails correction for the 10 hypotheses tested (needs $p < 0.005$). At 8 weeks the relationship weakens and loses significance. A hypothesis worth more data, not a confirmed signal.

RECOMMENDATION

Flag the anomaly. Don't trade the level.

1. Set the trigger

Basis beyond ± 2 std. dev. of the trailing 12-week mean ($\approx \pm \$2.15$ from a $-\$0.91$ baseline) triggers a manual review of pipeline/maintenance news — not an automatic trade.

2. Watch capacity, not just weather

Track pipeline capacity additions (like Matterhorn) as a structural leading indicator for basis narrowing.

3. Don't act on rig count yet

The relationship is directionally real but not confirmed at this sample size — a hypothesis for the next iteration, not today's signal.

WHAT I'D DO WITH MORE DATA OR TIME

This pre-empts the obvious follow-up questions

- Extend past Jan 21, 2026 — EIA discontinued this exact report; migrate the parser to its replacement (WNGSR Supplement) on EIA's Beta site
- Improve rig-count coverage beyond the current 64% by finding a source with denser structured data
- Add regional weather (HDD/CDD) as a third leading-indicator candidate alongside storage and rig count
- Wrap extraction in Airflow + dbt for a fully automated, scheduled pipeline

APPENDIX — DATA & LIMITATIONS

Source	What	Freq.
FRED API	Henry Hub spot price	Daily
EIA Weekly Update (parsed)	Waha price & rig count	Weekly
EIA API v2, South Central	Regional storage	Weekly

Known limitations

Waha coverage ends Jan 21, 2026 (EIA discontinued the report). Only 111 of 196 weeks have a usable Waha price — EIA only writes the parseable sentence in weeks of notable price moves. Storage/basis dates don't share a calendar day (Friday vs. Wednesday), joined via merge-as-of. Henry Hub (FRED) and Waha (parsed prose) use different underlying methodologies — directionally reliable, not penny-precise.